

Layout commands

These helpers make poster geometry explicit.

- `pframe`: one Beamer frame plus left/right gutters.
- `pcolumn{width}`: a top-aligned minipage of exact width.
- `prow`: puts columns on one row; use it for nested columns.
- `pblock[style]{title}{height}`: block with a target height.

Use `alert` or `example` as optional block styles. If content is taller than the target, the block grows instead of overprinting the next block.

Usual skeleton

`pframe` → `pcolumn` → `pblock`

Nested layout: `prow` → `pcolumn` → `pblock`.

Width recipe

Pick the number of columns first, then solve for widths.

- Top level: $(N + 1)s + Nw = \text{\texttt{\textbackslash textwidth}}$.
- Nested row: $\sum w_i + (N - 1)s = \text{\texttt{parent width}}$.
- Here: $s = \text{\texttt{\textbackslash pcolumnsepwidth}}$ and $w = \text{\texttt{\textbackslash colwidth}}$.

Why not Beamer columns?

Beamer's columns is made for slides, not grid-filling posters.

- It adds internal spacing, so the visible width is not just the number you typed.
- Column height follows content; it does not reserve an exact block height.
- Nested layouts are harder to make flush with the outer

Height recipe

Give every block a target height and let the last block fill the remainder.

$$\sum h_i + (M - 1) \text{\texttt{\textbackslash blocksepheight}} = \text{\texttt{\textbackslash textheight}}$$

For a column with blocks of 24cm, 12.5cm, and one remaining block:

$$\text{\texttt{\textbackslash textheight}} - 24\text{cm} - 12.5\text{cm} - 2\text{\texttt{\textbackslash blocksepheight}}$$

Use

Pattern

Same-height stack choose fixed cm values

Fill leftover space subtract used heights from `\textheight`

Two-column span $2\text{\texttt{\textbackslash colwidth}} + \text{\texttt{\textbackslash pcolumnsepwidth}}$

Too much content block grows; shorten text or increase height

Keep these formulas near the length definitions in the preamble.

Changing type size

Global sizes live in `beamerthemerpi.sty`.

- `block title`: usually `\Large` and bold.
- `block body`: usually `\large`.
- `headline title`: usually `\Huge`.

Local override: `{\Large ...}`. Exact size: `\fontsize{34}{40}\selectfont`.

Changing separator lengths

Two lengths control most whitespace.

- `\pcolumnsepwidth`: horizontal gutter between columns.
- `\blocksepheight`: vertical gap between stacked blocks.

Set them in the preamble:

Mini layout checklist

1. Choose poster size in `beamerposter`.
2. Choose `\pcolumnsepwidth`.
3. Solve for `\colwidth`.
4. Set span widths, e.g. `\twocolwidth`.
5. Pick block heights to sum to `\textheight`.
6. Build, then trim text or adjust heights.

Rule of thumb

Make the grid first, then write to fit it. Clean posters have shared gutters, aligned bottoms, and short text.

References

- [1] Claude E. Shannon.
A mathematical theory of communication.
Bell System Technical Journal, 27(3):379–423, 1948.